A geometric sequence has first term a and common ratio r

Given that $S_\infty = 3a$

(a) show that $r = \frac{2}{3}$

(2)

Given also that

$$u_2 - u_4 = 16$$

where u_k is the kth term of this sequence,

(b) find the value of S_{10} giving your answer to one decimal place.

(5)